Obtained from Ryerson's Mechanical Engineering Course Union (MECU) www.ryemecu.com

Ryerson University
Department of Mechanical and Industrial Engineering

MTL 200 – Materials Science Fundamentals

Midterm Test No.2

March 25, 2011

Instructor: Professor Daolun Chen, *Dr.rer.nat., Ph.D., P.Eng.*

INSTRUCTIONS

1. Duration is **50 minutes**.
2. This is a closed book test.
3. **Questions 1 and 2 are compulsory. Answer any two more questions from #3 to #5, and tick the answered questions at the top of next page.**
4. Back pages may be used if extra space is needed.
5. Please use an ink pen/ballpoint pen, and do not use a pencil (except for drawings if any).
6. Total marks: **40**

MECU

Questions	1	2	3	4	5	Total
Questions answered	√	√	✓		✓	
Maximum marks	10	10	10	10	10	40
Actual marks	7	4	5		3	19

MARKS: 10 −3

1. Indicate the correct answer by circling TRUE or FALSE.

(1) Liquidus line normally represents the boundary between a solid phase region and a two-phase region with the liquid [e.g., the boundary $\alpha/(\alpha+L)$].

TRUE / FALSE ✓

(2) A phase may be defined as a homogeneous portion of a system that has uniform physical and chemical characteristics.

TRUE / FALSE ✓

(3) The volume fraction of phases present in a phase diagram can be calculated using the lever rule.

TRUE / FALSE ✗

(4) Eutectic structure is a single-phase microstructure.

TRUE / FALSE ✓

(5) Peritectic reaction is an incongruent transformation.

TRUE / FALSE ✓

(6) In ceramic materials, cations are negatively charged nonmetallic ions.

TRUE / FALSE ✗

MECU

(7) Cesium chloride (CsCl) has an FCC-like crystal structure.

TRUE / FALSE

(8) Three primary polymorphic crystalline forms of silica (SiO_2) are quartz, cristobalite, and tridymite.

TRUE / FALSE

(9) Geometrical isomers are possible within mer units having a double bond between chain carbon atoms.

TRUE / FALSE

(10) Thermoplastic polymers are the polymeric materials that will soften when heated and harden when cooled. Such processes are totally reversible.

TRUE / FALSE

MARKS: 10 -6

2. Circle the correct answer of the following questions.

(1) Major factors affecting the microstructure of an alloy include

(a) alloying element, concentration, and heat treatment.

(b) mass fraction, concentration, and heat treatment.

(c) alloying element, mass fraction, and heat treatment.

(d) alloying element, concentration, and mass fraction.

(2) In the following reactions during equilibrium cooling (α, β, γ stands for solid phases), which is a eutectoid reaction?

(a) $L + \alpha \rightarrow \beta$.

(b) $L + \beta \rightarrow \gamma$.

(c) $L \rightarrow \alpha + \beta$.

(d) $\gamma \rightarrow \alpha + \beta$.

MECU

(3) In ceramic structures the coordination number of 3 for the cation corresponds to:

(a) tetrahedron.

(b) octahedron.

(c) planar equilateral triangle.

(d) linear manner.

(4) The most common coordination numbers for ceramic materials are

(a) 2, 3 and 4.

(b) 3, 4 and 6.

(c) 4, 6 and 8.

(d) 6, 8 and 12.

(5) From a practical perspective, the addition of the network modifiers (e.g., CaO, Na_2O) and intermediates (e.g., TiO_2, Al_2O_3) in the silica (SiO_2) glasses is used to

(a) increase the melting point and viscosity.

(b) increase the viscosity and density.

(c) decrease the melting point and viscosity.

(d) decrease the viscosity and density.

(6) Al_2O_3-Cr_2O_3 ceramic phase diagram is

(a) an isomorphous phase diagram.

(b) a eutectic phase diagram.

(c) a eutectoid phase diagram.

(d) a peritectic phase diagram.

(7) In polytetrafluoroethylene,

(a) all hydrogen atoms in polyethylene are replaced with F atoms.

(b) every second hydrogen atom in polyethylene is replaced with an F atom.

(c) every fourth hydrogen atom in polyethylene is replaced with an F atom.

(d) every eighth hydrogen atom in polyethylene is replaced with an F atom.

MECU

(8) Two possible types of configurations for stereoisomers are

(a) isotactic and atactic.

(b) cis and trans.

(c) cis and isotactic.

(d) trans and isotactic.

(9) The more irregular and random the mer arrangements, the ① the tendency for the development of ②.

(a) ① = smaller, ② = the degree of polymerization.

(b) ① = smaller, ② = noncrystallinity.

(c) ① = greater, ② = the degree of polymerization.

(d) ① = greater, ② = noncrystallinity.

(10) Most glass-forming operations are carried out within the following temperature range

(a) between the working and melting points.

(b) between the working and annealing points.

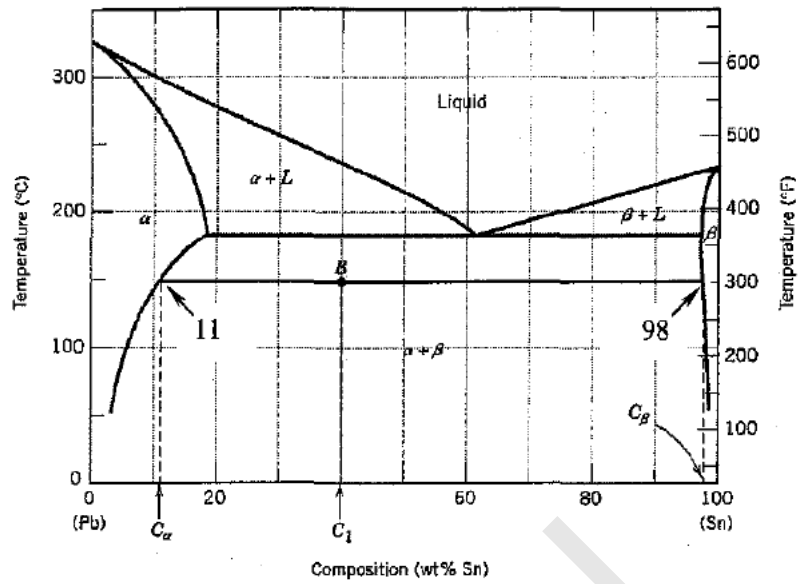
(c) between the working and softening points.

(d) between the working and strain points.

MECU

5

3. For a 40 wt% Sn-60 wt% Pb alloy at 150°C, calculate the relative amount of each phase present in terms of (a) mass fraction and (b) volume fraction. At 150°C take the densities of Pb and Sn to be 11.23 and 7.24 g/cm^3, respectively.



a) $W_\alpha = \frac{C_\beta - C_0}{C_\beta - C_\alpha}$

$= \frac{98 - 40}{98 - 11}$

$\boxed{W_\alpha = 0.67}$

$W_\beta = 1 - W_\alpha$

$\boxed{W_\beta = 0.33}$

2

2

b) $V_\alpha = \frac{\frac{0.67}{11.23}}{\frac{0.67}{11.23} + \frac{0.33}{7.24}}$

$\boxed{V_\alpha = 0.57}$

$V_\beta = 1 - V_\alpha$

1

$\boxed{V_\beta = 0.43}$

4. The unit cell for $MgFe_2O_4$ has cubic symmetry with a unit cell edge length of 0.836 nm. If the density of this material is 4.52 g/cm^3, calculate its atomic packing factor (The radius of Mg^{2+}, Fe^{3+} and O^{2-} is 0.072 nm, 0.069 nm and 0.140 nm, respectively).

MECU

5. The density and associated percent crystallinity for two nylon 6,6 materials are as follows:

3

ρ (g/cm³)	crystallinity (%)
1.188	67.3
1.152	43.7

(a) Calculate the densities of totally crystalline and totally amorphous nylon 6,6.
(b) Determine the density of a specimen having 55.4% crystallinity.

a) $\% = 67.3 = \frac{\rho_c(\rho_s - \rho_a)}{\rho_s(\rho_c - \rho_a)} \times 100$

$0.673 = \frac{\rho_c(\rho_s - \rho_a)}{\rho_s(\rho_c - \rho_a)}$

$\frac{(0.673)(1.188(\rho_c - \rho_a))}{\rho_c} - 1.188 + \rho_a = 0$

$\frac{0.8\rho_c - 0.8\rho_a}{} = \cancel{-1.188 + \rho_a = 0}\ (1.188 - \rho_a)(\rho_c)$

$0 = 0.388\rho_c - \rho_a\rho_c + 0.8\rho_a$

see back

$$\frac{(0.437)(1.152(P_c - P_a))}{P_c} = 1.152 - P_a$$

$$0.5P_c - 0.5P_a = 1.152P_c - P_cP_a$$

$$0 = 0.652P_c - P_cP_a + 0.5P_a$$

$$\cancel{0.388P_c - P_cP_a + 0.8P_a = 0.652P_c - P_cP_a + 0.5P_a}$$

$$0.652P_c = (0.5 - P_c)(P_a)$$

$$P_a = \frac{0.652P_c}{0.5 - P_c}$$

$$0.388P_c - P_c\left(\frac{0.652P_c}{0.5 - P_c}\right)$$

in (a), both P_c & P_a can be calculated.

---

(6) ?